The documentation and process conversion measures necessary to comply with this document shall be completed by 19 December, 2001.

INCH-POUND
MIL-PRF-19500/295C
19 September 2001
SUPERSEDING
MIL-S-19500/295B
5 October 1992

PERFORMANCE SPECIFICATION

* SEMICONDUCTOR DEVICE, FIELD-EFFECT TRANSISTORS, P-CHANNEL, SILICON TYPES 2N2608, JAN, AND UB

Inactive for new design for the 2N2608 device after 19 September 2001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for P-channel, junction, silicon field-effect transistors. One level of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-18) and figure 2.

1.3 Maximum ratings.

| Types | $P_T$ (1) $T_A$ = +25°C | $V_{GSS}$ | $T_{STG}$ and $T_{OP}$ |
|---|---|---|---|
| | mW | V dc | °C |
| 2N2608 | 300 | 30 | -65 to +200 |

(1) Derate linearly, 1.71 mW/°C for $T_A$ = +25°C.

1.4 Primary electrical characteristics at $T_A$ = +25°C.

| Limit | $I_{DSS}$ $V_{DS}$ = -5 V dc $V_{GS}$ = 0 | $V_{GS(off)}$ $V_{DS}$ = -5 V dc $I_D$ = -1.0 µA dc | $C_{ISS}$ $V_{DS}$ = 3 V dc $V_{GS}$ = 0 V dc f = 1 MHz | $\lvert Y_{FS} \rvert$ $V_{DS}$ = -5 V dc $V_{GS}$ = 0 V dc f = 1 kHz | $I_{GSS}$ $V_{GS}$ = 15 V dc $V_{DS}$ = 0 V dc |
|---|---|---|---|---|---|
| | mA dc | V dc | pF | µmho | nA dc |
| Minimum | -1.0 | 0.75 | | 1,000 | |
| Maximum | -5.0 | 6.00 | 10 | 4,500 | 7.5 |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2.

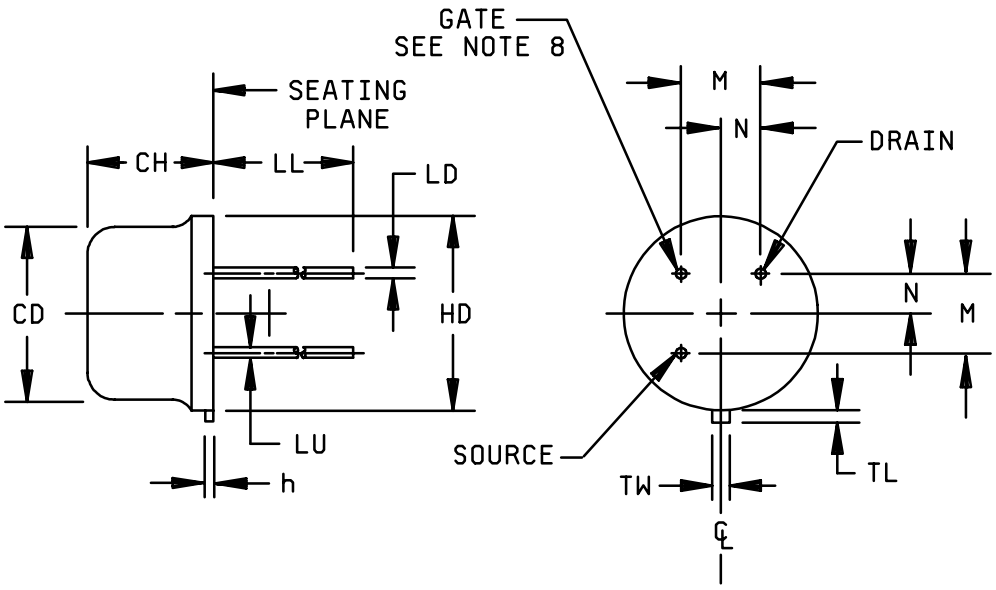
3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3.

3.6 Electrical test requirements. The electrical test requirements shall be group A as specified herein.



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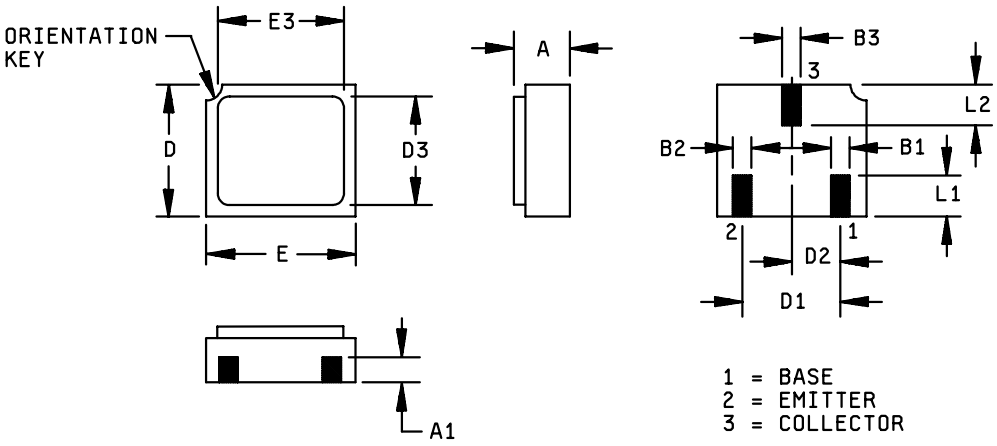

| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | |
| CH | .170 | .210 | 4.32 | 5.33 | |
| HD | .209 | .230 | 5.31 | 5.84 | |
| h | | .020 | | 0.51 | |
| LD | .016 | .021 | 0.41 | 0.53 | 2, 7 |
| LL | .500 | .750 | 12.70 | 19.05 | 7 |
| LU | .016 | .019 | 0.41 | 0.48 | 3,7 |
| M | .0707 Nom | | 1.80 Nom | | 4 |
| N | .0354 Nom | | 0.90 Nom | | 4 |
| TL | .028 | .048 | 0.71 | 1.22 | 6 |
| TW | .036 | .046 | 0.91 | 1.17 | |

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gauging plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 3 shows the preferred measurement method.
5. The gate shall be electrically connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All three leads. (see 3.4.1)
8. Diameter of leads in this zone is not controlled.

FIGURE 1. Physical dimensions.



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .046 | .056 | 1.17 | 1.42 |
| A1 | .017 | .035 | 0.43 | 0.89 |
| B1 | .016 | .024 | 0.41 | 0.61 |
| B2 | .016 | .024 | 0.41 | 0.61 |
| B3 | .016 | .024 | 0.41 | 0.61 |
| D | .085 | .108 | 2.16 | 2.74 |
| D1 | .071 | .079 | 1.80 | 2.01 |
| D2 | .035 | .039 | 0.89 | 0.99 |
| D3 | .085 | .108 | 2.16 | 2.74 |
| E | .115 | .128 | 2.92 | 3.25 |
| E3 | | .128 | | 3.25 |
| L1 | .022 | .038 | 0.56 | 0.96 |
| L2 | .022 | .038 | 0.56 | 0.96 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

* FIGURE 2. Physical dimensions, surface mount (UB version).



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3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.3.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIb JAN of MIL-PRF-19500 and 4.3.2.1 herein. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table II herein.

4.3.2.1 Group B inspection, table VIb, of MIL-PRF-19500.

a. Subgroup 3, conditions for steady-state operation life are as follows: $V_{CB} \geq 10$ V dc, $T_A = +150°C$, $V_{DS} = 0$; $V_{GS} = 24$ V dc.

b. Subgroup 4, condition for bond strength is condition A.

c. Subgroup 5, not applicable.

d. Subgroup 6, condition for high-temperature life (nonoperating) is $T_{STG} = +200°C$.

4.3.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table II herein.

4.3.3.1 Group C inspection, table VII, of MIL-PRF-19500.

a. Subgroup 2, conditions for terminal strength is test condition A.

b. Subgroup 5 and 7, not applicable.

c. Subgroup 6, condition for steady-state operation life (accelerated) are as follows: $T_A = +150°C$, $V_{DS} = 0$; $V_{GS} = 24$ V dc.



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TABLE I. Group A inspection .

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Gate - source breakdown voltage | 3401 | Bias condition C; $I_G$ = 1.0 µA dc; $V_{DS}$ = 0. | $V_{(BR)GSS}$ | 30 | | V dc |
| Gate reverse current | 3411 | Bias condition C; V = 30 $V_{GS}$ dc; $V_{DS}$ = 0. | $I_{GSS1}$ | | 10 | nA dc |
| Gate reverse current | 3411 | Bias condition C; V = 15 $V_{GS}$ dc; $V_{DS}$ = 0. | $I_{GSS2}$ | | 7.5 | nA |
| Zero-gate-voltage drain current | 3413 | Bias condition C; $V_{DS}$ = 5 V dc; $V_{GS}$ = 0; pulsed. | $I_{DDSS1}$ | -1.0 | -5.0 | mA dc |
| Gate to source cutoff voltage | 3403 | $I_D$ = 1 µA dc; $V_{DS}$ = 5 V dc | $V_{GS(off)}$ | 0.75 | 6.0 | V dc |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_A$ = +125°C | | | | |
| Drain current | 3413 | Bias condition C, $V_{GS}$ = 0 V dc; $V_{DS}$ = 5 | $I_{DDSS2}$ | -0.56 | | mA dc |
| Gate reverse current | 3411 | Bias condition C; $V_{GS}$ = 15 V dc; $V_{DS}$ = 0. | $I_{GSS3}$ | | 15 | µA dc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Small-signal, common-source, short-circuit, forward transfer admittance | 3455 | $V_{DS}$ = 5 V dc; $V_{GS}$ = 0; f = 1 kHz. | $\lvert Y_{fs1} \rvert$ | | 4,500 | µmho |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Small-signal common-source, short-circuit, forward transfer admittance | 3455 | $V_{DS}$ = 5 V dc; $V_{GS}$ = 0; f = 1 kHz. | $\lvert Y_{fs2} \rvert$ | 1,000 | 4,500 | µmho |
| Small-signal common-source, short-circuit, input capacitance | 3431 | $V_{DS}$ = 5 V dc; $V_{GS}$ = 0; f = 1 MHz. | $C_{iss}$ | | 10 | pF |
| Common-source spot noise figure | | $V_{DS}$ = 5 V dc; $V_{GS}$ = 0; $B_W$ = 16 percent; $R_G$ = 1 Mohm; $e_{gen}$ = 1.82 mV; $R_L$ = 470 ohms; f = 1 kHz; (see figure 4). | NF | | 3 | dB |
| Subgroups 5, 6, and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.



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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1 | Gate reverse current | 3411 | Bias condition C; $V_{GS}$ = 15 V dc; $V_{DS}$ = 0 | $I_{GSS1}$ | | 8 | nA dc |
| 2 | Gate reverse current | 3411 | Bias condition C; $V_{GS}$ = 15 V dc; $V_{DS}$ = 0 | $I_{GSS2}$ | | 15 | nA dc |
| 3 | Drain current | 3413 | Bias condition C; $V_{GS}$ = 5 V dc; $V_{DS}$ = 0 | $I_{DSS1}$ | -1.0 | -5.0 | mA dc |
| 4 | Drain current | 3413 | Bias condition C; $V_{GS}$ = 5 V dc; $V_{DS}$ = 0 | $I_{DSS2}$ | -0.90 | -5.5 | mA dc |
| 5 | Small-signal, common-source short-circuit, forward transfer admittance | 3455 | $V_{DS}$ = 5 V dc; $V_{GS}$ = 0; f = 1 kHz | $\lvert Y_{fs1} \rvert$ | 1,000 | 4,500 | µmho |
| 6 | Small-signal, common-source short-circuit, forward transfer admittance | 3455 | $V_{DS}$ = 5 V dc; $V_{GS}$ = 0; f = 1 kHz | $\lvert Y_{fs3} \rvert$ | 900 | 4,950 | µmho |

1/ The electrical measurements for table VIb (JAN) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, steps 1, 3 and 5.
   b. Subgroups 3 and 6, see table II herein, steps 2, 4 and 6.

2/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
   a. Subgroups 2 and 3, see table II herein, steps 1, 3 and 5.
   b. Subgroup 6, see table II herein, steps 2, 4 and 6.



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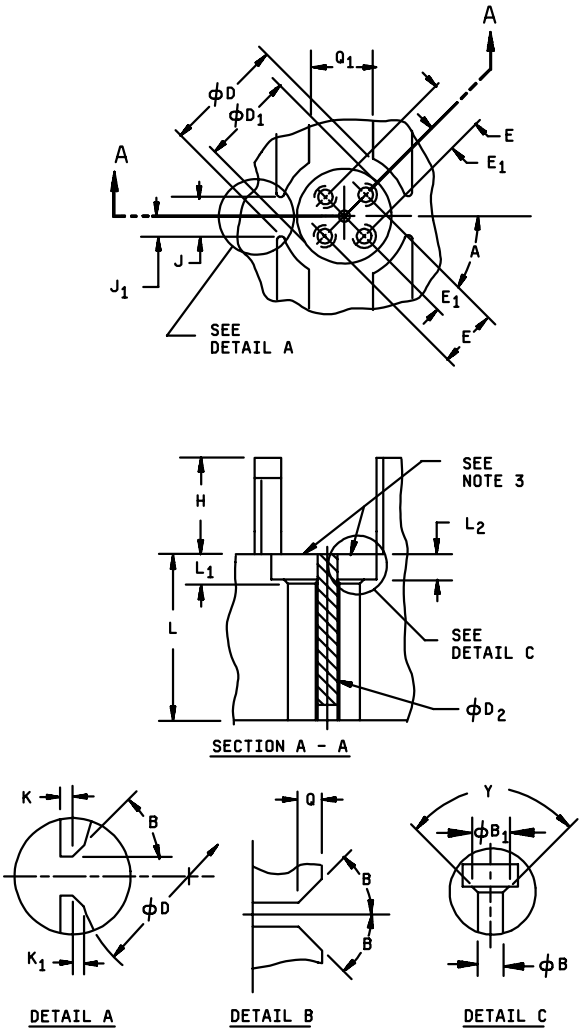



FIGURE 3. Gauge for lead and tab location for transistor type 2N2608.



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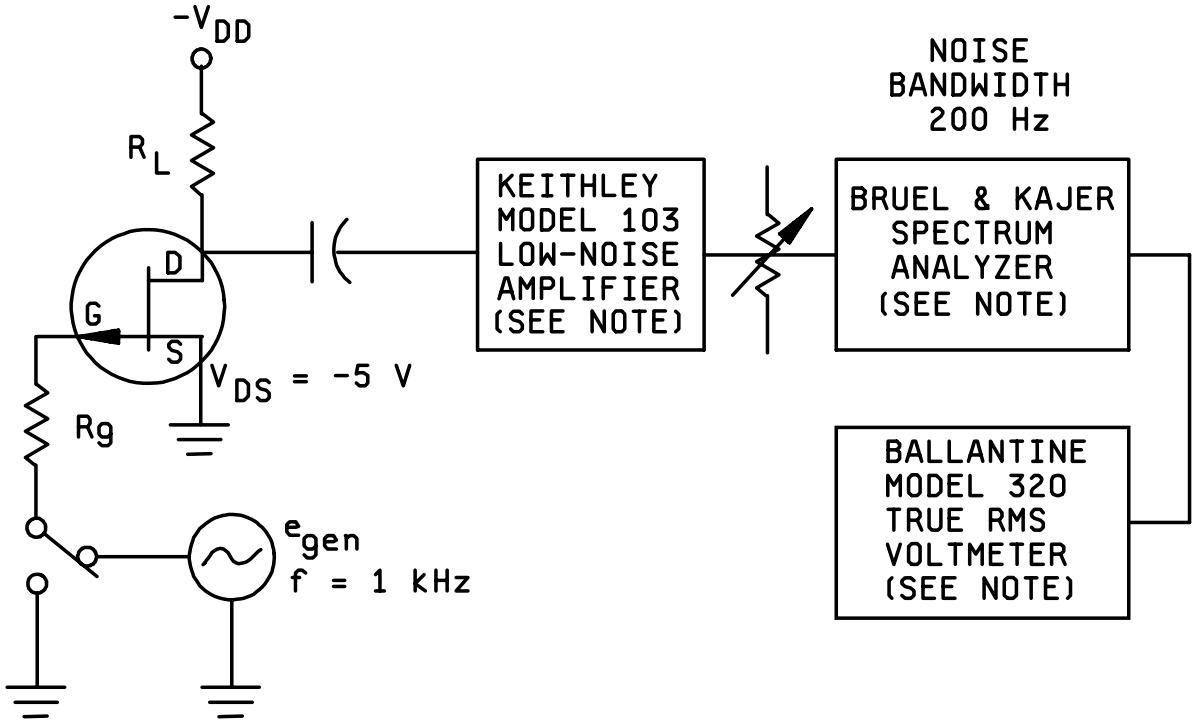
| Symbols | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| $\phi B$ | .0325 | .0335 | 0.83 | 0.85 | 4 |
| $\phi B_1$ | .043 Nom | | 1.09 Nom | | 4 |
| $\phi D$ | .2310 | .2315 | 5.87 | 5.88 | |
| $\phi D_1$ | .159 | .161 | 4.04 | 4.09 | |
| $\phi D_2$ | .040 Nom | | 1.02 Nom | | 5 |
| E | .0995 | .1005 | 2.53 | 2.55 | |
| $E_1$ | .0495 | .0505 | 1.26 | 1.28 | |
| H | .145 | .155 | 3.68 | 3.94 | |
| J | .0470 | .0475 | 1.19 | 1.21 | |
| $J_1$ | .0235 | .0245 | 0.60 | 0.62 | |
| K | .009 | .011 | 0.23 | 0.28 | |
| $K_1$ | .005Nom | | .127 Nom | | |
| L | .372 | .378 | 9.45 | 9.60 | |
| $L_1$ | .054 | .055 | 1.37 | 1.40 | |
| $L_2$ | .043 Nom | | 1.09 Nom | | |
| Q | .040 Nom | | 1.02 Nom | | |
| $Q_1$ | .123 | .127 | 3.12 | 3.23 | |
| A | 44.90° | 45.10° | | | |
| B | 45° Nom | | | | |
| Y | 90° Nom | | | | |

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. The following gauge procedure shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) +.010 inch (0.25 mm) from seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not to be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 inch (0.03 mm).
4. Four holes.
5. Pressed in.

FIGURE 3. Gauge for lead and tab location for transistor type 2N2608 - Continued.



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NOTE: Or equivalents

Procedure: The voltage and current shall be applied to the terminals, and the noise figure shall then be measured as follows:

1. Connect egen to the input of the device under test.
2. Adjust gain of system to give 0 dB reading on a convenient scale of the Ballantine voltmeter or equivalent.
3. Switch the device input to ground.
4. Increase system gain by 60 dB.
5. Read noise figure directly in dB's on Ballintine scale or equivalent.

FIGURE 4. Common-source spot noise figure test circuit.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see 5.1).

d. Lead finish (see 3.4.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - NW
Air Force - 11
DLA - CC

Review activities:
Army - AR
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2485)



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**STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL**

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| **I RECOMMEND A CHANGE:** | 1. DOCUMENT NUMBER<br>MIL-PRF-19500/295C | 2. DOCUMENT DATE<br>19 September 2001 |
|---|---|---|

3. **DOCUMENT TITLE** SEMICONDUCTOR DEVICE, FIELD-EFFECT TRANSISTORS, P-CHANNEL SILICON, TYPES 2N2608, JAN, AND UB.

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>COMMERCIAL<br>DSN<br>FAX<br>EMAIL | 7. DATE SUBMITTED |

8. PREPARING ACTIVITY

| a. Point of Contact<br>Alan Barone | b. TELEPHONE<br>Commercial<br>614-692-0510 | DSN<br>850-0510 | FAX<br>614-692-6939 | EMAIL<br>alan.barone@dscc.dla.mil |
|---|---|---|---|---|
| c. ADDRESS<br>Defense Supply Center Columbus<br>ATTN: DSCC-VAC<br>P.O. Box 3990<br>Columbus, OH 43216-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman, Suite 2533<br>Fort Belvoir, VA 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 | | | |

**DD Form 1426, Feb 1999 (EG)** Previous editions are obsolete WHS/DIOR, Feb 99

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